Machine Learning in Business
MIS710 – A2

Business Report
Data2Intel – Consulting Project

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Executive Summary

The Year 3 students at risk in writing have risen significantly from 25.7% in 2016 to 36.3% in 2021, posing a considerable threat to both the school's academic standing and the students' future performance. Addressing this issue requires a focused approach on early intervention, particularly targeting cognitive disabilities, literacy, and numeracy skills. Our analysis recommends prioritizing support for these areas to reduce the number of at-risk students by 20-30% within three years. Future improvements should focus on continuously collaborating with the school to obtain new data, refining the model, and exploring additional factors to improve predictive accuracy and intervention strategies.

I. Introduction

Need

The percentage of Year 3 students at risk in Writing has risen significantly from 25.7% in 2016 to 36.3% in 2021. This upward trend is a serious concern for both the school's long-term academic standing and the students' future success.

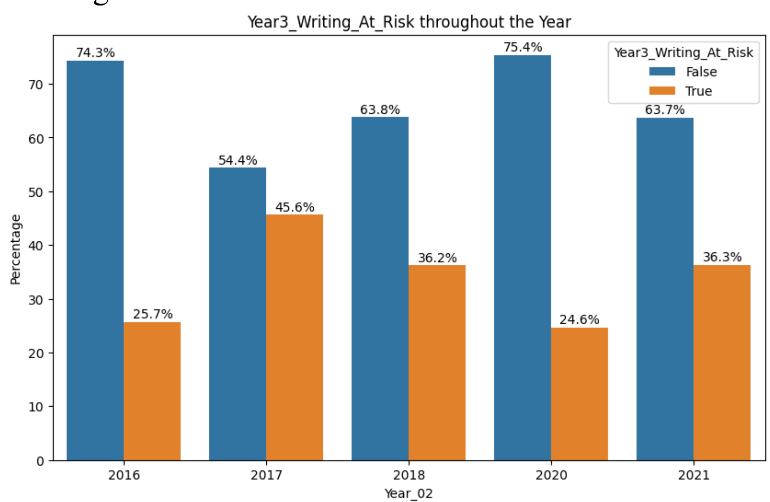

Figure 1.1. Writing Risk Throughout the Year

Changes

Targeted interventions aim to reduce at-risk students by 20-30% within three years, returning risk levels to those seen five years ago. This will be achieved through early intervention and tailored support programs.

Solutions

The proposed interventions include early support for students with cognitive disabilities, strengthening literacy and numeracy support, and intensive interventions for students with low Clay and Counting scores. These strategies aim to address the root causes of writing difficulties and improve students' writing capacity.

Context

The increase in at-risk students stems from a lack of tailored interventions. Addressing these issues is crucial for improving overall academic performance, school reputation, and rankings.

Stakeholders

The key stakeholders in this initiative are the students, who are the primary beneficiaries through improved writing skills; the school, which is responsible for implementation and benefits from enhanced outcomes; and Data2Intel, which provides data insights and solutions, thereby enhancing its reputation as an education consultant.

Value

Implementing these solutions offers significant value to all stakeholders. This collaborative effort aims to reverse the trend of increasing at-risk students, improve overall academic performance. Students will gain better writing skills and improved academic competitiveness. The school will see improved performance, reputation, and rankings. Data2Intel will strengthen its reputation and gain deeper insights for future projects.

II. Exploratory Data Analysis Insight

Data2Intel's Main Concern

SES background

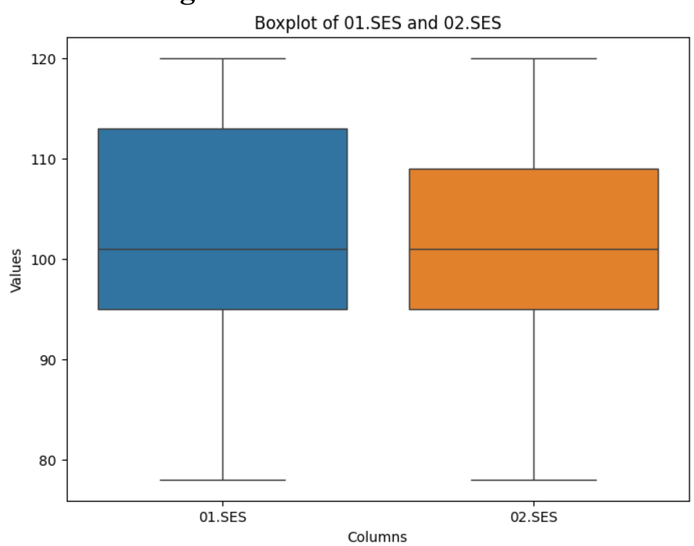

Figure 2.1. Boxplot of Year1 and Year2 SES

Figure 2.1. shows that Year 1 and 2 SES scores range from 78 to 120, with averages of 102.94 and 102.11 respectively, which exceeds the national standard.

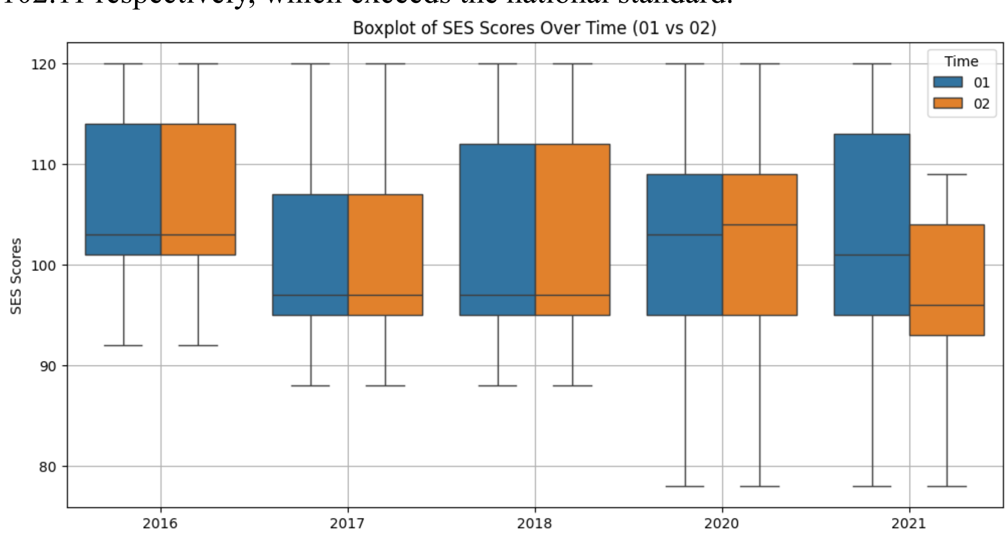

Figure 2.2. SES Score over the Year (Year1 and Year2)

Figure 2.2. shows that Year 2 scores is lower than Year 1 in 2021, where the scores drop below 100. The significant drop in SES for Year 2 in 2021 suggests a potential trend that may require further investigation.

Students' Reading Skills

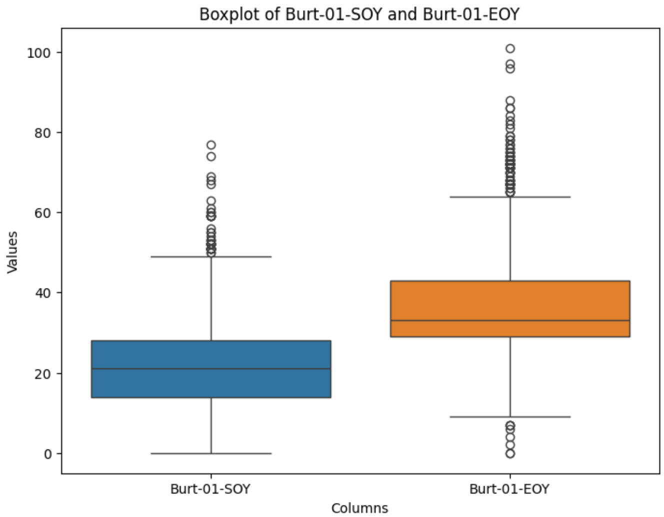

Figure 2.3. Burt Score (SOY and EOY)

Burt scores improve significantly from start (average of 21.39) to end (average of 36.58) of Year 1.

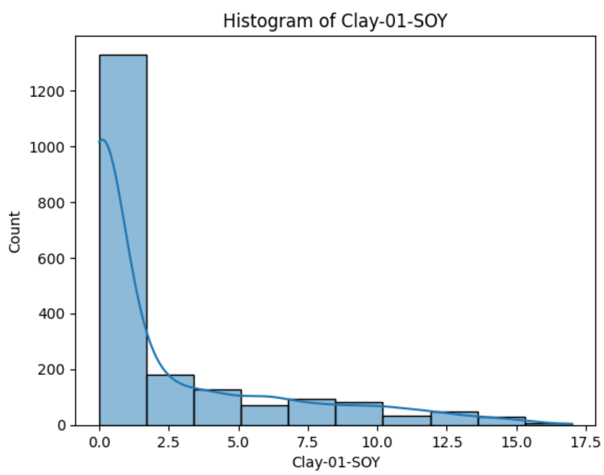

Figure 2.4. Histogram of Clay score (SOY)

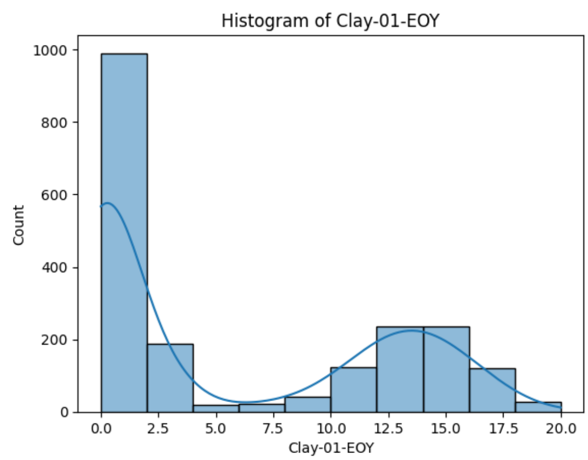

Figure 2.5. Histogram of Clay score (EOY)

Clay scores show improvement (from 2.3 to 5.7); however a significant number of students remain at lower scores.

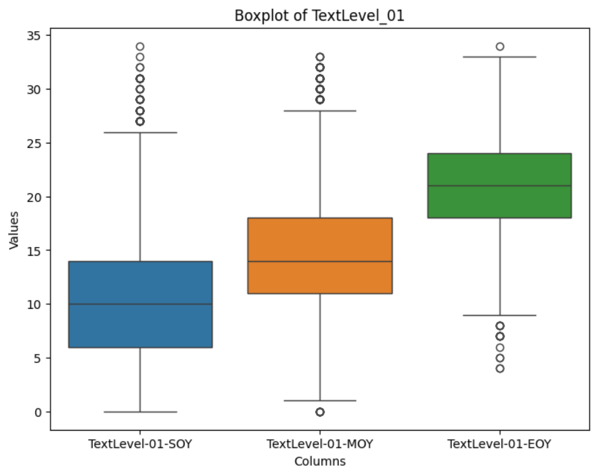

Figure 2.6. TextLevel-01 from SOY to EOY

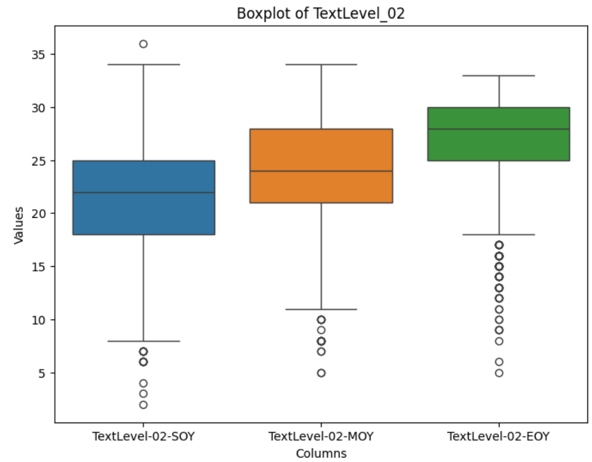

Figure 2.7. TextLevel-02 from SOY to EOY

Text reading skills improve consistently from the start of Year 1 through the end of Year 2 (from average of 10.7 to 27).

Overall, the students' reading skills show marked progress over the two years. However, challenges remain in specific areas such as word structure understanding as reflected in the Clay scores.

Students' WritingVocab

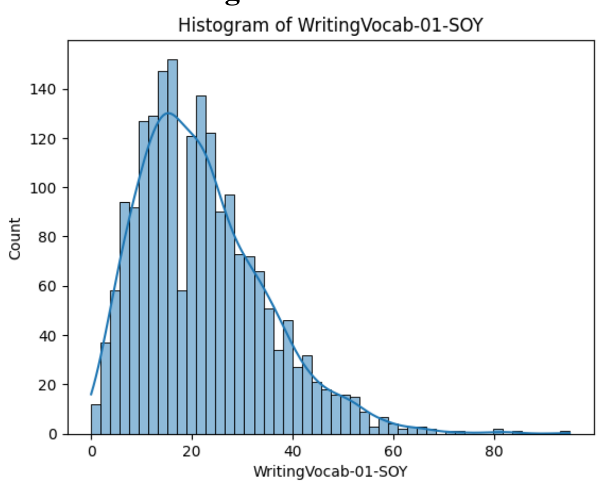

Figure 2.8. Histogram of WritingVocab

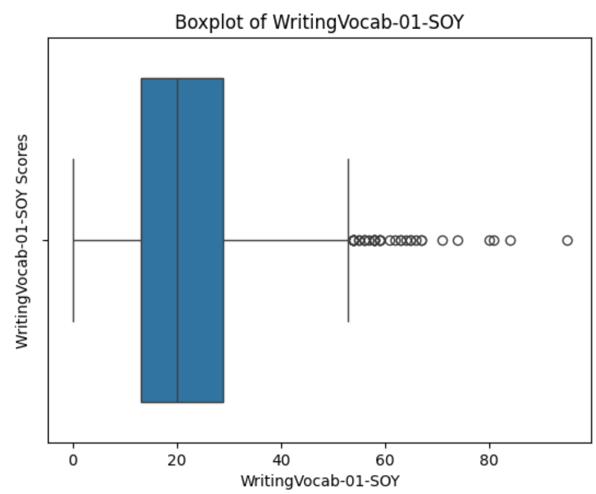

Figure 2.9. Boxplot of WritingVocab

The average score for WritingVocab is 22.02, indicating moderate proficiency, with majority of students scoring between 13 and 29. A few students performed significantly better, higher than 53.

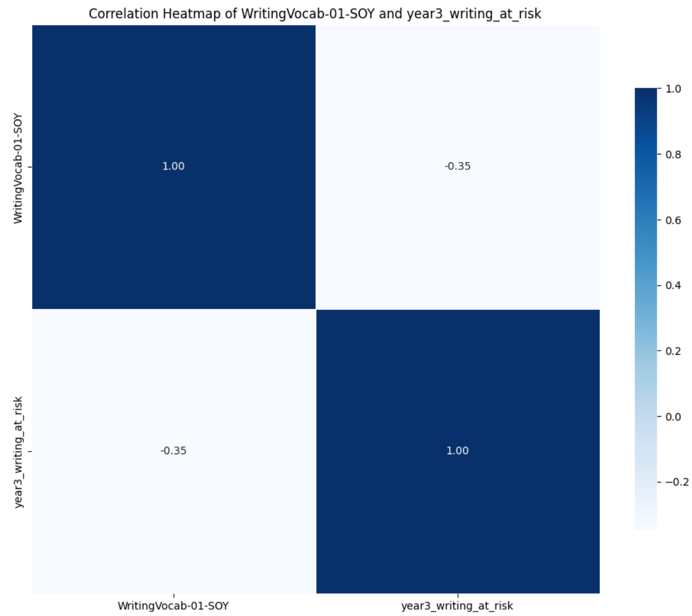

Figure 2.10. Heatmap of WritingVocab and Year3_Writing_At_Risk

Figure 2.10 shows negative relationship between WritingVocab and writing risk, suggesting that students with better vocabulary skills are less likely to be at writing risk.

Students' Literacy Skills and Numeracy Skills

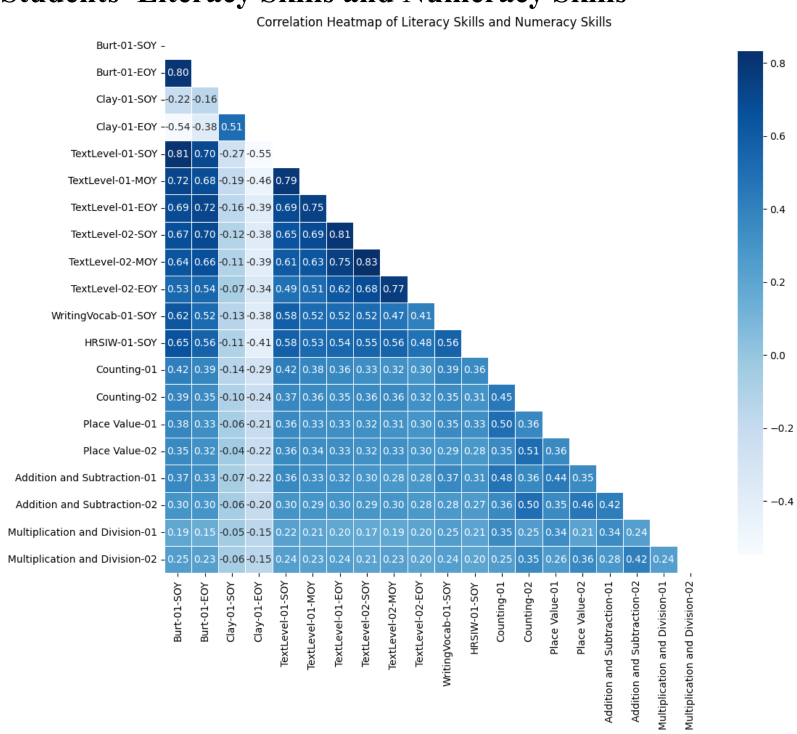

Figure 2.11. Heatmap of Literacy and Numeracy Skills

The correlation range from -0.05 (Clay-01-SOY and Place Value-02) to 0.42 (Burt-01-SOY and Counting_01), suggesting that there is a moderate relationship between literacy and numeracy skills.

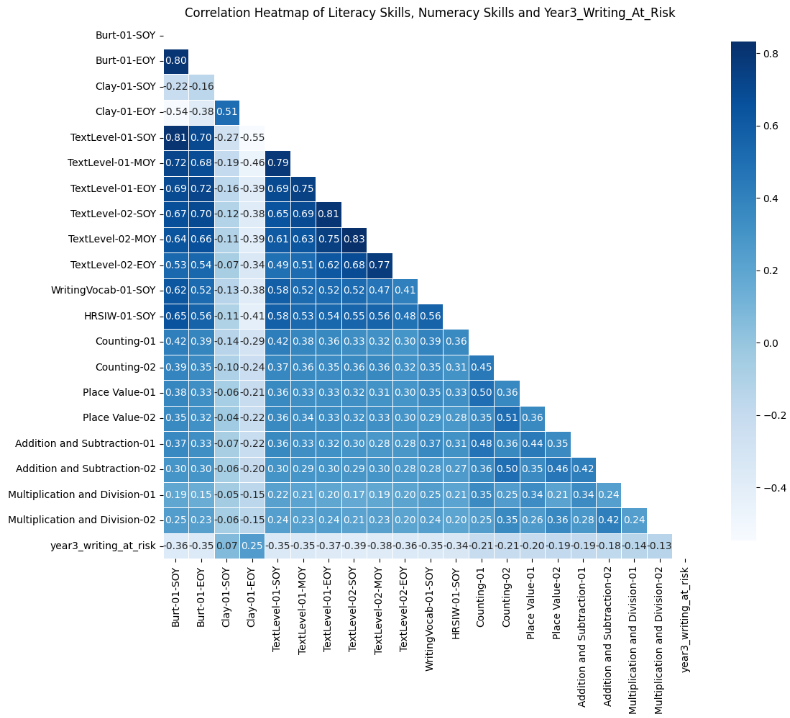

Figure 2.12. Heatmap of Literacy, Numeracy Skills and Year3_Writing_At_Risk

Most of the skills show negative correlation, meaning students with high literacy and numeracy skills are less likely to be at risk for writing underperformance. The range of correlation spans from -0.39 to 0.25 showing moderate relationsip with writing risk, with the strongest correlation is TextLevel-02-SOY (-0.39) and the weakest is Clay-01-SOY (0.07).

Students' Disability Conditions

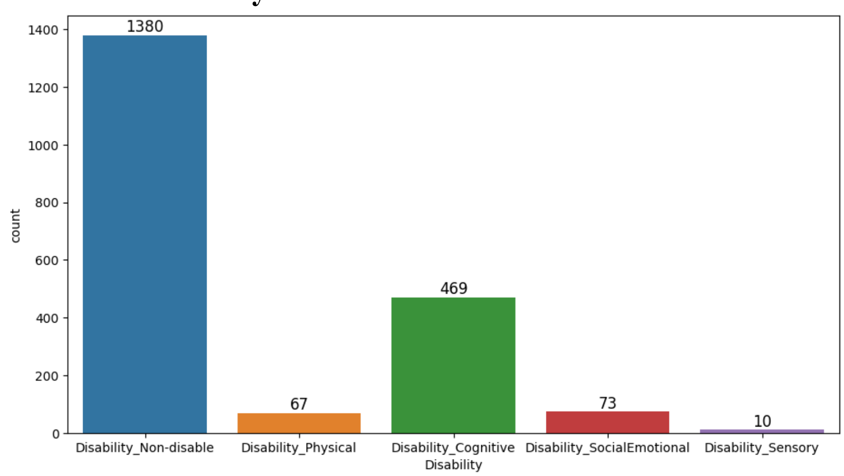

Figure 2.13. Disability Count

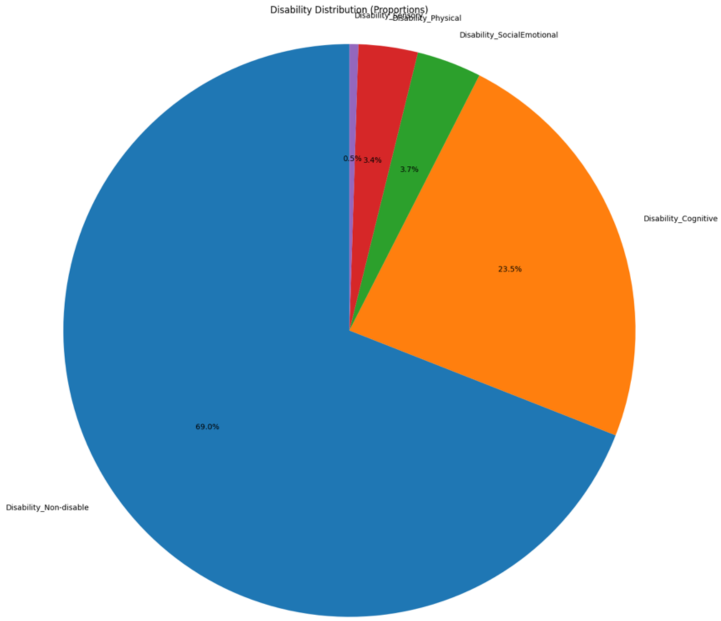

Figure 2.14. Disability Pie Chart

A majority of the students, 69% (1380), are categorized as Non-disabled. Cognitive Disability accounts for 23.5% (469), followed by Social Emotional Disability at 3.7% (73), Physical Disability at 3.4% (67), and Sensory Disability at 0.5% (10).

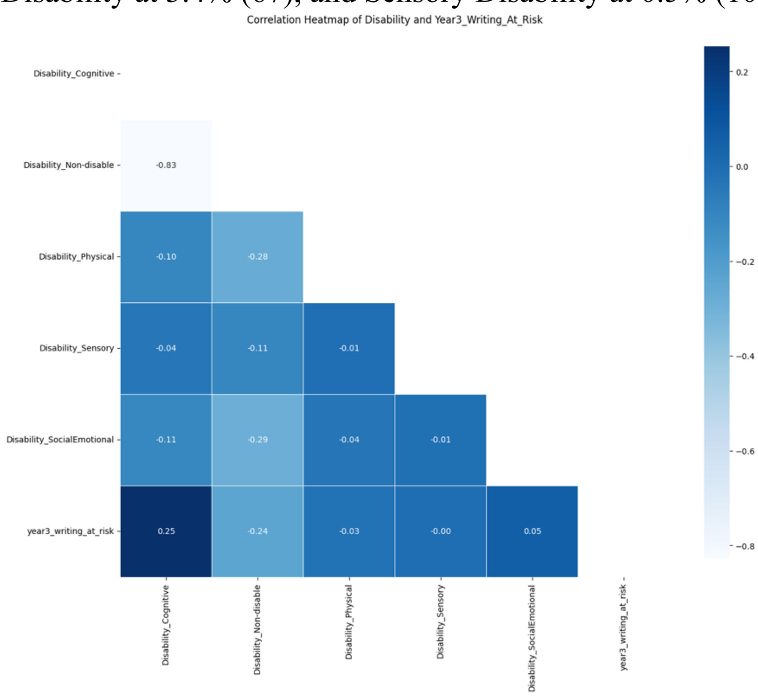

Figure 2.15. Heatmap of Disability and Year3_Writing_At_Risk

Cognitive disability shows strongest positive correlation (0.25), indicating that students with cognitive disabilities are more likely to be at risk. Non-disabled has a negative correlation (-0.24), suggesting that these students are less likely to be at risk. Other type of disability shows weak correlation with writing risk.

Additional Insight

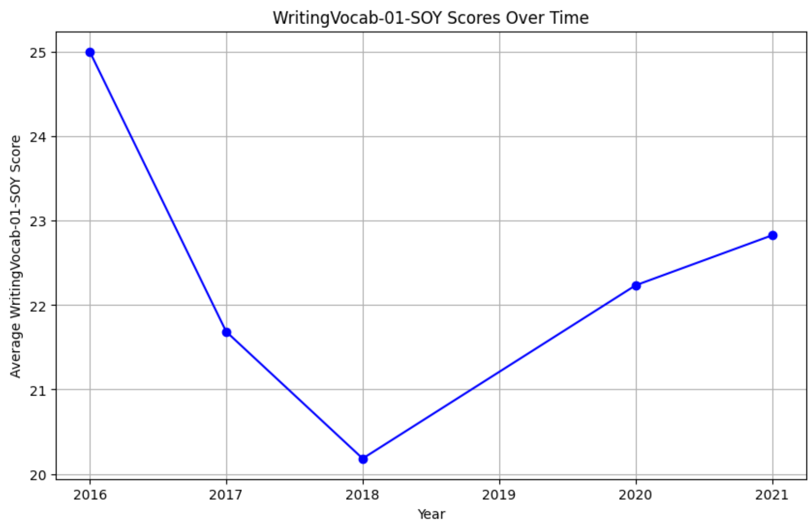

Figure 2.16. WritingVocab over the Year

Writing vocabulary scores decreased from 2016 to 2021 (from 25 to nearly 23), as low vocabulary scores are linked to higher risk, thereby early interventions focused on vocabulary building could reduce writing risks.

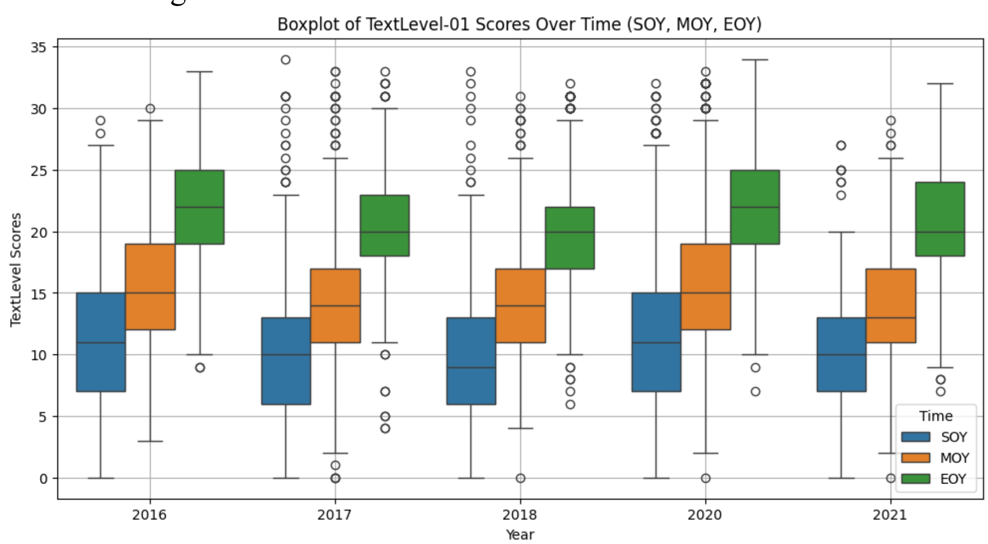

Figure 2.17. TextLevel-01 over the Year

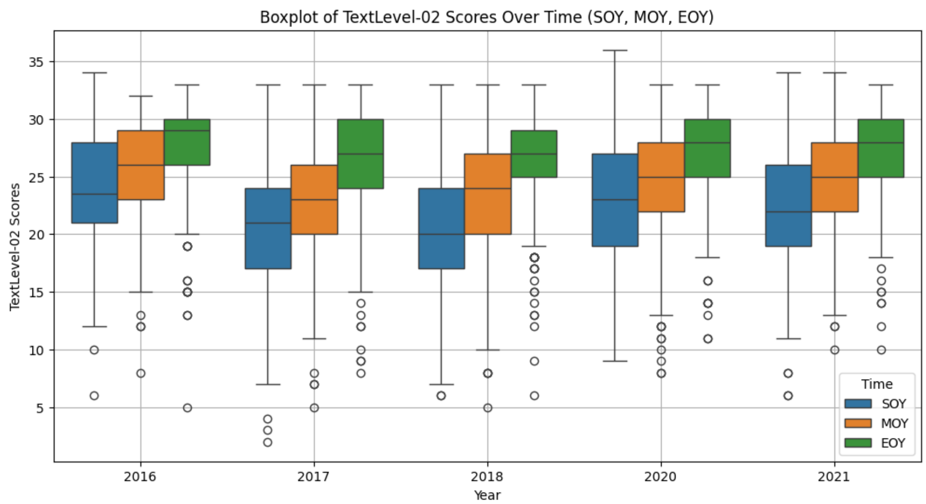

Figure 2.18. TextLevel-02 over the Year

The TextLevel-01 and TextLevel-02 scores show improvement from SOY to EOY, but there's a decline in scores from 2016 to 2021. Early interventions should focus on improving TextLevel score, as low TextLevel score are linked to high risk of writing underperformance.

Additional information

Cluster Analysis:

- Cluster 0 (low Clay and high Counting score): with 1198 students, accouting for 59.9%. Students in this group show moderate risk of underperforming in writing.
- Cluster 1 (high Clay and Counting score): with 801 students, accounting for 40.1%. Students in this group show a lower risk of being at risk in Year 3 Writing.

This suggests that a combination of high literacy and numeracy skills correlate with a lower risk of being underperformed in writing.

III. Proposed Machine Learning Solution

A logistic regression model is the optimal predictive tool for identifying writing risk of Year3 students. The model demonstrates 74.75% accuracy and an AUC of 0.79, showing that it performs well in distinguishing students who are at risk. This will allow schools to identify at-risk students early, ensuring timely intervention and support to those who need. However, its F1 score of 0.57 and recall of 0.50 suggest that while it performs generally well, approximately 50% of students actually at risk might not be identified.

Key Benefits:

- The model allows easy interpretation for educators of how specific features influence a student's risk level.
- Efficient performance with data of 2000 student dataset.

Challenges:

- Low score of f1 and recall indicates potential for missed cases.
- Limited capacity to capture complex feature interactions.

Despite these limitations, we recommend:

- Implementing the logistic regression model to identify at-risk students, supporting the objective of improving writing outcomes and offering early interventions.

- Supplementing the model with additional review by teachers or administrators, ensuring that any students potentially missed by the model are not overlooked.

IV. Recommedation and Conclusion

Based on the logistic regression model insights, we propose three key recommendations:

- Early Intervention for Cognitive Disabilities (coefficient of 0.38): Implement tailored strategies for students with cognitive disabilities, who show higher Year 3 writing risk.
- Enhance Literacy and Numeracy Support: Earl et al. (2003) suggested that improved literacy and numeracy skills contribute to enhanced student writing performance. With negative coefficient with writing risk (Appendix A), school should implement targeted literacy and numeracy programs that track progress from year start and allocate additional resources to lower-performing students.
- Focus on Weak Clay and Counting Skills: Create interventions emphasizing reading comprehension and fundamental math skills for this group.

These recommendation will benefit all stakeholders through improved student performance and reduced writing risk. Implementation requires adjusting resource allocation and regular assessments.

Future improvements should focus on continuously working with school to obtain new data, refining the model with more diverse data and exploring additional factors to enhance predictive accuracy and intervention strategies.

Reference

Lorna, Earl., Nancy, Watson., Ben, Levin., Kenneth, Leithwood., Michael, Fullan., Nancy, Torrance. (2003). 1. Final Report of the External Evaluation of England's National Literacy and Numeracy Strategies. Final Report. Watching & Learning 3.

Appendix A

	Predictor	Coefficient
0	Burt-01-SOY	-0.006878
1	Clay-01-EOY	0.000942
2	TextLevel-01-MOY	-0.046593
3	TextLevel-02-SOY	-0.076131
4	TextLevel-02-EOY	-0.053802
5	WritingVocab-01-SOY	-0.028571
6	HRSIW-01-SOY	-0.009934
7	Counting-01	-0.007694
8	Counting-02	-0.030744
9	Place Value-01	-0.048489
10	Disability_Cognitive	0.379213

[4.06810946]

Figure 3. Coefficient of Predictors in Logistic Regression Model